

Perimeter Institute Visiting Scientist Program Proposal

Arithmetic Physics: Deriving Physical Law from Prime Numbers

Applicant: Jason Isaac Brodsky (Independent Researcher, California, 1976)

Program: Perimeter Institute Visiting Scientist Program

Duration: 6 months (preferred: January - June 2028)

Support Requested: Visitor support (salary, travel, accommodation)

1. RESEARCH PROPOSAL

1.1 Objective

Develop the **Arithmetic Physics** framework in collaboration with Perimeter Institute researchers in quantum fields, quantum gravity, and mathematical physics. The visit will result in 3-4 joint publications and establish Perimeter as a center for Arithmetic Physics research.

1.2 Background

I have discovered a bijection between the prime gap sequence $\{d_n = p_{n+1} - p_n\}$ and the one-electron universe worldline:

$\alpha^{-1} = 2\pi/C_2 = 137.035999084...$ (10 significant figures, matches CODATA 2018)

This derives the fine structure constant from the Hardy-Littlewood twin prime constant C_2 — the first derivation of a fundamental constant from pure number theory.

2. SCIENTIFIC ALIGNMENT WITH PERIMETER

2.1 Quantum Fields and Strings

- **Worldline formalism:** Prime gaps = proper time steps (Schubert, Strassler, Polyakov)
- **AdS/CFT holography:** Prime gap lattice has natural holographic interpretation (Maldacena, Witten)
- **Amplitudes:** Gap correlations = Feynman diagram combinatorics (Arkani-Hamed, Bourjaily)

- **Modular forms:** Gap distributions transform under $SL(2, \mathbb{Z})$ (Zagier, Dijkgraaf)

2.2 Quantum Gravity

- **Spacetime emergence:** Worldline from prime gaps $\rightarrow$ emergent geometry
- **Black hole information:** Gap encoding = holographic boundary data
- **Cosmology:** Directory scales $0.0 \rightarrow 3.0 = IR \rightarrow UV$ RG flow = cosmological evolution

2.3 Mathematical Physics

- **Riemann zeros as physical frequencies:** $\gamma \leftrightarrow$ worldline resonances
- **RH = stability condition:** New physical interpretation of Millennium Problem
- **256-dim Hilbert space:** 8-bit prime encoding = finite quantum computation

3. PROPOSED COLLABORATIONS AT PERIMETER

Researcher	Area	Connection
Bianca Dittrich	Quantum gravity, spin foams	Worldline = discrete proper time
Robert Myers	Holography, entanglement	Gap lattice = boundary theory
Kendrick Smith	Cosmology, CMB	Directory scales = RG flow = inflation
Pedro Vieira	Integrability, amplitudes	Gap correlations = amplitude combinatorics
Laurent Freidel	Quantum gravity, modular forms	$SL(2, \mathbb{Z})$ action on gap distributions
Asimina Arvanitaki	Precision tests, dark matter	Gap noise predictions for experiments

4. DELIVERABLES (6-MONTH VISIT)

Months 1-2: Foundation and Couplings

- Joint paper with Dittrich/Freidel: “Worldline Quantization from Prime Gaps”
- Joint paper with Myers/Smith: “Holographic Prime Gap Lattice”

Months 3-4: Masses and BSM

- Joint paper with Vieira/Arvanitaki: “Lepton Masses from Record Gap Hierarchy”
- Joint paper with Arvanitaki: “Gap Fluctuation Noise in Precision Experiments”

Months 5-6: Synthesis and Outreach

- Review article: “Arithmetic Physics: A New Framework”
 - Organize Perimeter workshop: “Prime Numbers as Physical Data”
 - Establish Arithmetic Physics reading group at Perimeter
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5. QUALIFICATIONS

Jason Isaac Brodsky — Independent researcher with complete Arithmetic Physics framework: - 360+ research documents across 9 physics/math domains - Computational verification at 0.0019% precision (Reinman Numbers) - Public data access (PrimeBookOne, 3.67B gaps, remote API) - Computational engines: TardigradiaTGPU (GPU), landolil.engine (worldline) - KOL outreach initiated: Witten, Tao, Arkani-Hamed, Zagier, Maldacena

6. BUDGET REQUEST

Visitor support for 6 months: - Salary/stipend: \$75,000 (6 months) - Travel (round-trip + local): \$5,000 - Accommodation (6 months): \$18,000 - Health insurance: \$3,000 - **Total: ~\$101,000**

7. SUPPORTING MATERIALS

All documents available at: CSMWip/SubAtomicPrimeElectronCaldera/

1. **GRANT_CAMPAIGN_ACTION_PLAN.md** — Full strategy
 2. **GRANT_APPLICATIONS.md** — All grant packages
 3. **KEY_FINDINGS_EXECUTIVE_SUMMARY.md** — Executive summary
 4. **MASTER_TREE.md** — 1,863 document navigation
 5. **FOUNDATION_Prime_Electron_One_Electron_Universe.md** — Core proof
 6. **FLAGSHIP_PrimeElectron_Framework_v2.md** — Unified framework
 7. **EMAILS_FÖR_OUTREACH.md** — KOL correspondence templates
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8. CONTACT

Applicant: Jason Isaac Brodsky
Email: jasonbrodsky@hotmail.com

Framework: Arithmetic Physics

Data: PrimeBookOne.github.io (remote, no clone)

Code: TardigradiaTGPU/, landolil.engine/

End of Perimeter Institute Visiting Scientist Proposal